

Michael Cassidy

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EDUCATION

Queen's University

Sept 2022 – May 2027

Bachelor of Applied Sciences in Engineering, Mechatronics and Robotics

Kingston, ON

- Recipient of Schulich Leader Scholarship valued at \$80 000
- Queen's Engineering Competition - Senior Design - 1st Place
- GPA: 3.84, Dean's Scholar

PROFESSIONAL EXPERIENCE

Mechanical Engineer Intern

May – Aug 2026

Canadian Strategic Missions Corporation

Waterloo, ON

- Independently led design and delivery of a representative prototype reactor assembly for a Department of National Defense stakeholder presentation, producing **GD&T**-compliant technical drawings and coordinating directly with machinists to procure and manufacture components in a fast-paced, resource-constrained startup environment.
- Modeled the kinematics of a control rod drive mechanism in **CAD** and created **dynamics simulation** to size mechanical components and design a **custom electromagnetic actuator**, including magnetic **circuit design** and coil winding specification, meeting load and response requirements.
- Designed the mechanism to operate through a pressure boundary in compliance with nuclear **safety standards**.
- Defined **design-for-manufacturing** (DFM) requirements and researched applicable **ASME** Boiler and Pressure Vessel Codes for a mass-manufacturable micro-reactor core and automated assembly line, informing manufacturability-driven design decisions early in development.

Software Engineer Intern

May – Aug 2023, 2024, 2025

Teranet

Toronto, ON

- Led development of 12 Java REST APIs in 2024, reducing the average request response time by over a minute for key operations compared to the legacy system, significantly improving efficiency.
- Developed a new integrated application in 2025, cutting response times by over 100ms across all requests and reducing operating costs by unifying functionality from multiple legacy systems.
- Enhanced codebase maintainability and team collaboration by standardizing design practices and improving readability across applications.

EXTRACURRICULARS

Team Captain

Apr 2025 - Present

Queen's Aerospace Design Team

Kingston, ON

- Led and integrated design efforts across **structures, propulsion, aerodynamics, and manufacturing** for a 50-person AIAA team, resulting in an airplane design which achieved all of the team's technical and performance objectives.
- Designed the aircraft structure and composite skin, performing **carbon fiber layups** and designing lightweight, mission-specific mechanisms to meet the competition's yearly design challenge requirements, achieving a total aircraft weight under 5kg through continuous weight-driven design trade-offs.
- Developed and executed subsystem and full-aircraft **test programs** to validate the design ahead of competition, ensuring flight reliability.

Team Captain

Sept 2022 - May 2025

Queen's VEXU Robotics Team

Kingston, ON

- Led mechanical, electrical, and software development across an 80-person team, contributing hands-on across the full robotics stack: mechanical design, embedded electronics, and **guidance/navigation/control** (GNC) software, to build optimized, reliable competition robots.
- Designed several custom gearbox architectures, including multistage planetary and cycloidal designs, to maximize power density and reliability within tight space and weight constraints compared to competitor robots.
- Designed a **custom PCB** interfacing multiple sensors, including encoders, IMUs and LiDARs as part of a GNC system, enabling precise robot localization and control.
- Led mechanical development of four robots over three years, engineering **gear trains, pneumatic actuators, linkages, and manipulators** optimized for speed, precision, and reliability.
- Achieved **VEX AI Skills World Champion** and the Build Award at Worlds, recognized for durability, reliability, and manufacturing quality.